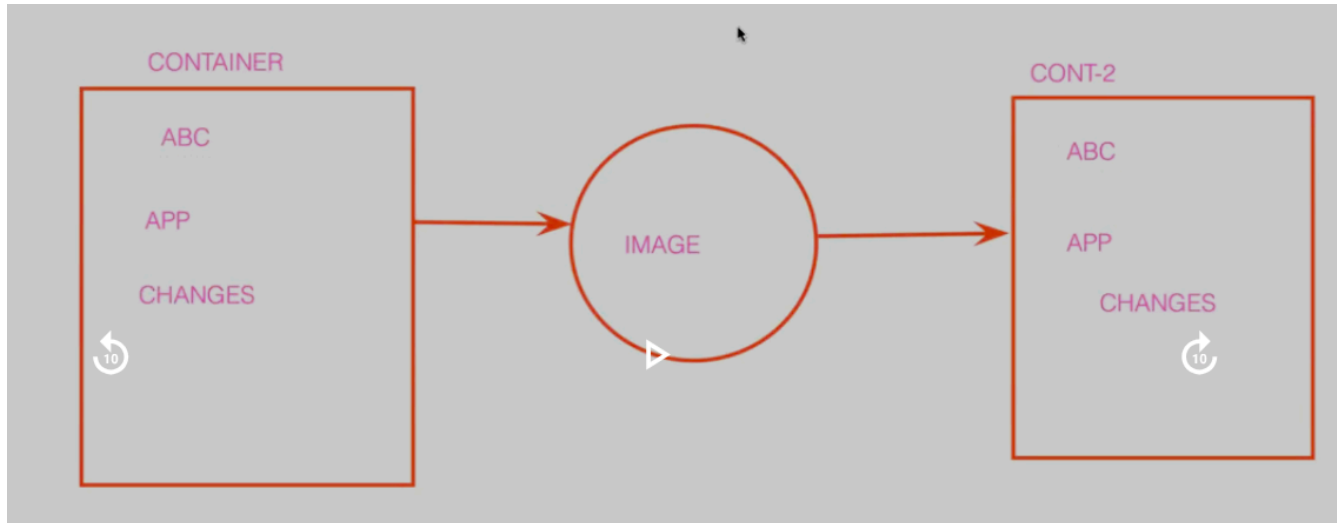


docker_class_6(49)

Docker volumes:

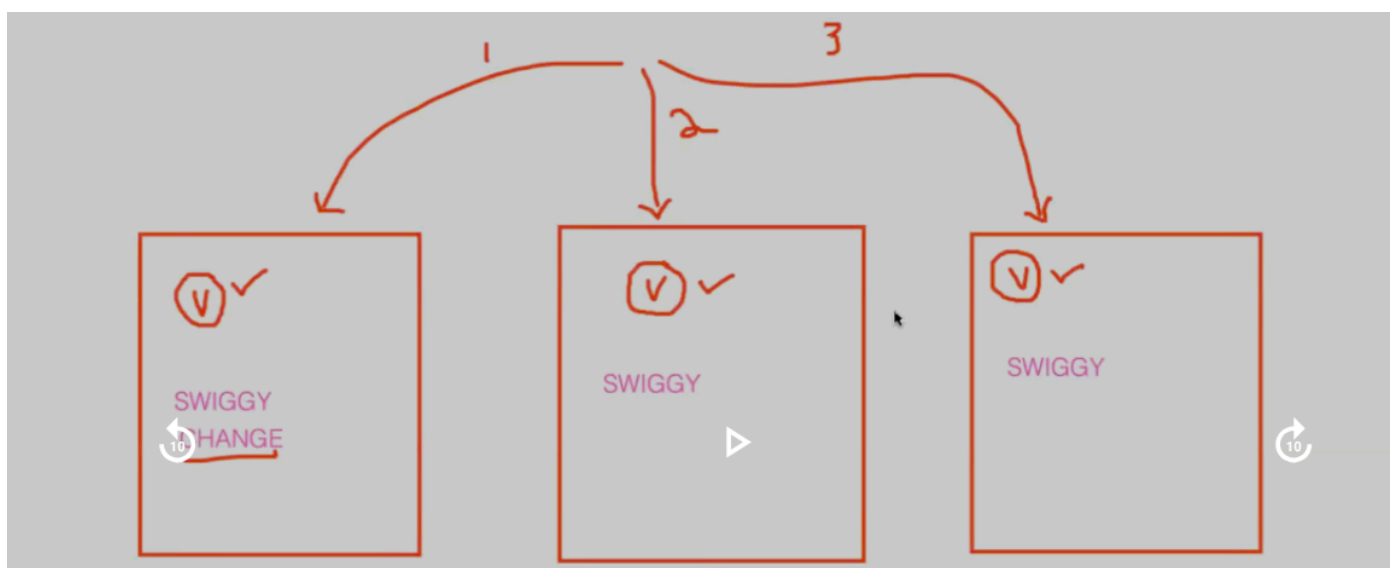
at what situation volumes will be used will see:



in cont1 it is running application same cont1 if i create one more like view as earlier we did cont1 --> create image--> through the image i will create container 2 but in cont1 if update is it replicate the changes in cont2 ans: is no or in cont2 if i update is it come cont1 ans: no so data replicate we use docker volume

docker volume is used to store the data inside the container and also outside the host machine it will replicate the data into any container

aim is if cont1 i update it should update cont2 also



if person1 is access cont1

if person2 access cont2

if person3 access cont3

in cont1 if i change the data only person 1 only access the data

and again person is access cont2 does he get latest data so every container we have to manual same data if we want so fot that if i use docker volume we can replciate the data into cont2 and cont3 v is volume.

DOCKER VOLUMES:

- When we create a container, a volume will be created.
- A volume is simply a directory inside our container.
- First, we have to declare the directory as a volume and then share the volume.
- Even if we stop/delete the container, we can still access the volume.
- You can declare a directory as a volume only while creating the container.
- We can't create a volume from an existing container.
- You can share one volume across multiple containers.
- A volume will not be included when you update an image.
- If Container-1 shares a volume with Container-2, changes made by Container-2 will also be available in Container-1.

Volume Mapping Methods:

1. Container ↔ Container

2. Host ↔ Container

- Volume = **storage managed by Docker**

- Stored on **host**, not inside container filesystem

- Volume is created **only if you specify it**

- Otherwise → container uses temporary filesystem (deleted later)

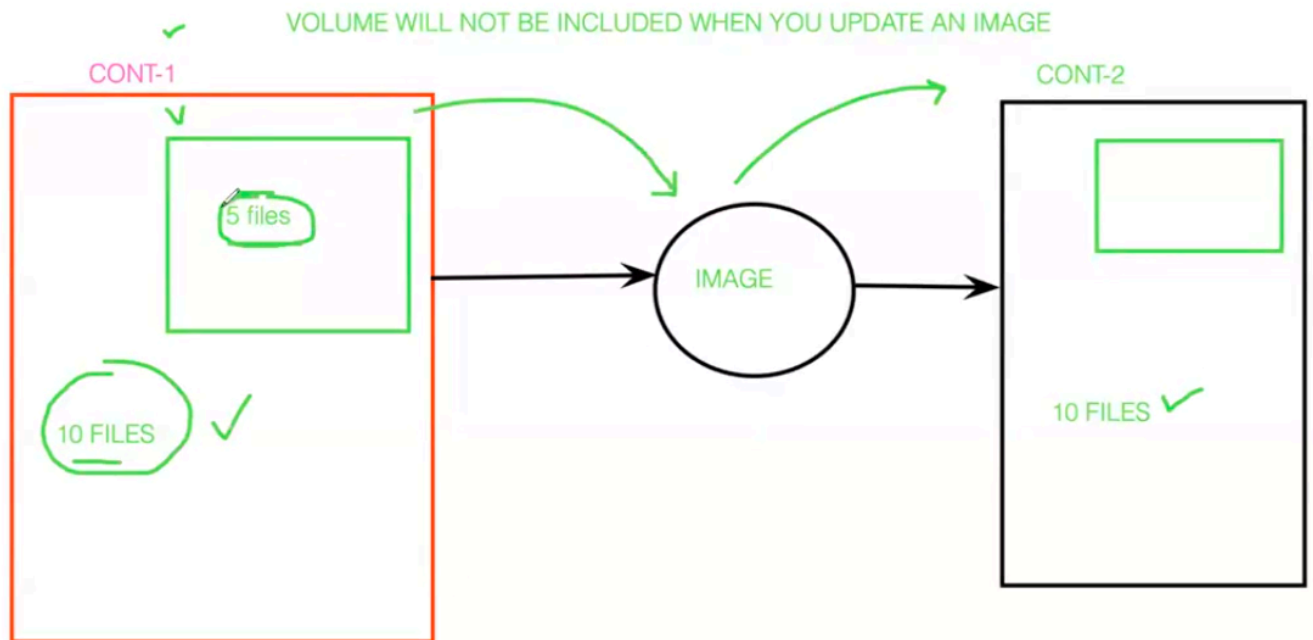
- You can create volume anytime:

- You can create and attach volumes later using:

- `docker run`

- `docker-compose`

- But you can't magically extract existing container filesystem into volume without manual steps



docker volume we can create 2 ways one is docker file and another is command way `docker run -it --name harish1 -v /FLM-Volume ubuntu`

\\[944181c2fe576c0d1b51375302e15865.png\\](\\[:/606378c361e34034a131346c8af8ed99\\](:/606378c361e34034a131346c8af8ed99%29 ":/606378c361e34034a131346c8af8ed99")) this volume looks like folder

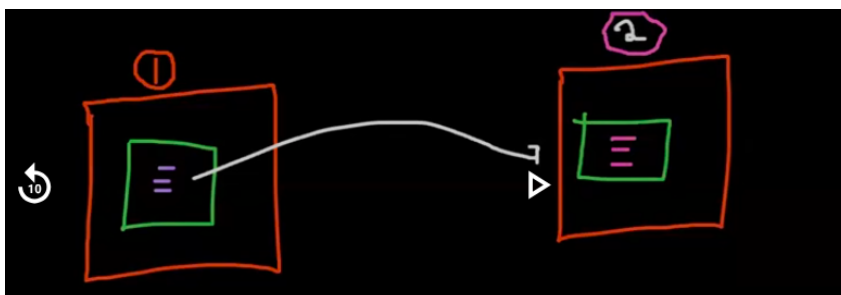
here i will create one folder `mkdir FLM-Folder` FLM-Folder we can remove but FLM-Volume we can not remove inside the container and 2. volume una data vere container share chesukuntam iam creating 3 files `CD FLM-Volume touch aws gcp azure` !\\

[cabac6b74412e551f8442bc9a0fa3646.png\\](\\[:/f8518f5689fc45e5932072ba6eb16b91\\](:/f8518f5689fc45e5932072ba6eb16b91%29 ":/f8518f5689fc45e5932072ba6eb16b91")) so this

container harish1 is sharable to other container if i want to share container 2 first before that in cont1 while run con1 you should have to declare then 3 files will come if want to access cont1 file in the

command needs to mention while create a container

i.e `\\*\\*docker run -it --name harish2 --privileged=true --volumes-from harish1 ubuntu\\*\\*`



```
root@3f79702be2ce:/FLM-Volume# ll
total 8
drwxr-xr-x 2 root root 4096 Mar 26 19:44 ./
drwxr-xr-x 1 root root 4096 Mar 26 19:54 ../
-rw-r--r-- 1 root root    0 Mar 26 19:44 aws
-rw-r--r-- 1 root root    0 Mar 26 19:44 azure
-rw-r--r-- 1 root root    0 Mar 26 19:44 gogle
root@3f79702be2ce:/FLM-Volume#
```

cont1 committing the container \--privileged means : shareble volume simply: sharable volume
from cont1 in container 2 iam able to see container 1 volume data but here no data now here i go
to vim mode

container 2 get container1 volume as you saw this i want to insert the data before that need
dependenc

apt update

apt install -y vim

vim aws

ha bro your great effort

in container 2 i kept a data

```
root@3f79702be2ce:/FLM-Volume# cat aws
hey bro your good bro do good be good
root@3f79702be2ce:/FLM-Volume#
```

now i exit from this container and enter container one i.e harish1

```
root@3f79702be2ce:/FLM-Volume# harish@aold2402006:~/docker$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
3f79702be2ce   ubuntu   "/bin/bash"             11 minutes ago Up 11 minutes           harish2
cb419056314c   ubuntu   "/bin/bash"             31 minutes ago Up 31 minutes           harish1
harish@aold2402006:~/docker$ docker attach harish1
root@cb419056314c:/FLM-Volume# cat aws
hey bro your good bro do good be good
root@cb419056314c:/FLM-Volume#
```

if you observer container 1 data is replicated

and here container 1 i will remove 2 files rm -rf azure google
so if am remove from cont1 automatically it will reflect into cont2

as ealier we discussed this point :

- o
- A volume will not be included when you update an image.
image will not update when you update image

this will try now

in container i want to attach some data

docker attach cont-2

```
root@3f79702be2ce:/FLM-Volume# ll
total 12
drwxr-xr-x 2 root root 4096 Mar 27 11:22 ./
drwxr-xr-x 1 root root 4096 Mar 26 19:54 ../
-rw-r--r-- 1 root root  38 Mar 26 20:03 aws
-rw-r--r-- 1 root root   0 Mar 27 11:22 file1
-rw-r--r-- 1 root root   0 Mar 27 11:22 file2
-rw-r--r-- 1 root root   0 Mar 27 11:22 file3
root@3f79702be2ce:/FLM-Volume#
```

inside volume i have created 3 files

now out side the container i will 10 files

```
root@3f79702be2ce:/FLM-Volume# cd
root@3f79702be2ce:~# touch container-files{1..10}
root@3f79702be2ce:~# ll
total 24
drwx----- 1 root root 4096 Mar 27 11:24 ./
drwxr-xr-x 1 root root 4096 Mar 26 19:54 ../
-rw----- 1 root root 296 Mar 27 03:00 .bash_history
-rw-r--r-- 1 root root 3106 Apr 22 2024 .bashrc
-rw-r--r-- 1 root root 161 Apr 22 2024 .profile
-rw----- 1 root root 760 Mar 26 20:03 .viminfo
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files1
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files10
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files2
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files3
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files4
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files5
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files6
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files7
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files8
-rw-r--r-- 1 root root 0 Mar 27 11:24 container-files9
root@3f79702be2ce:~#
```

total 14 files

exit from cont2

create image from cont2

docker commit harish2 my_image

```
harish@aold2402006:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
centos-httpd:latest	4d7a7ce59855	219MB	0B	U
cycle:latest	2115038f259f	170MB	0B	U
flm:2	89f44d20e45f	117MB	0B	U
flm:3	ae6677a3ebc5	117MB	0B	U
image_cr:latest	556a1e055d47	210MB	0B	U
imagelayer:latest	a1298eb677b5	245MB	0B	
my_image:latest	41eb7fd61720	210MB	0B	
myimage:latest	2062f55a4b8c	245MB	0B	U
python3:latest	c20d34b0e0e1	161MB	0B	U

Now from this image i will create container i.e

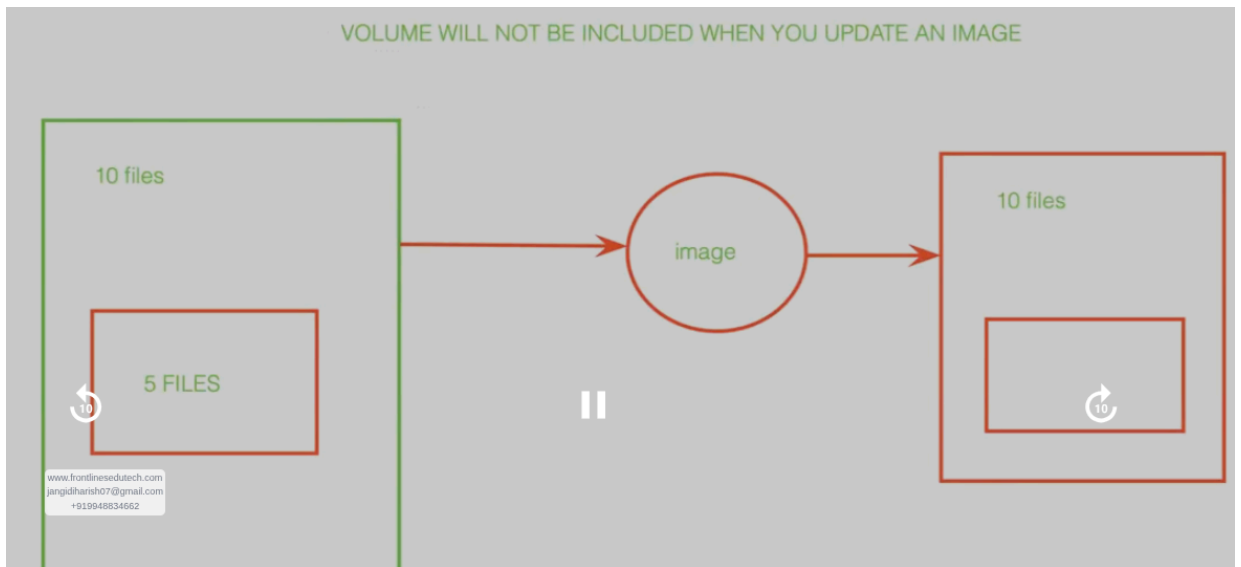
docker run -it --name cont3 my_image

```

harish@aold2402006:~$ docker run -it --name cont3 my_image
root@d8c4820eaf12:/# ll
total 60
drwxr-xr-x  1 root root 4096 Mar 27 11:39 ./
drwxr-xr-x  1 root root 4096 Mar 27 11:39 ../
-rwxr-xr-x  1 root root   0 Mar 27 11:39 .dockerenv*
drwxr-xr-x  2 root root 4096 Mar 26 19:54 FLM-Volume/
lrwxrwxrwx  1 root root   7 Apr 22  2024 bin -> usr/bin/
drwxr-xr-x  2 root root 4096 Apr 22  2024 boot/
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files1
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files10
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files2
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files3
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files4
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files5
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files6
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files7
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files8
-rw-r--r--  1 root root   0 Mar 27 11:31 container-files9
drwxr-xr-x  5 root root  360 Mar 27 11:39 dev/
drwxr-xr-x  1 root root 4096 Mar 27 11:39 etc/
drwxr-xr-x  3 root root 4096 Feb 17 02:09 home/
lrwxrwxrwx  1 root root   7 Apr 22  2024 lib -> usr/lib/
lrwxrwxrwx  1 root root   9 Apr 22  2024 lib64 -> usr/lib64/
drwxr-xr-x  2 root root 4096 Feb 17 02:02 media/
drwxr-xr-x  2 root root 4096 Feb 17 02:02 mnt/
drwxr-xr-x  2 root root 4096 Feb 17 02:02 opt/
dr-xr-xr-x 473 root root   0 Mar 27 11:39 proc/
drwx----- 1 root root 4096 Mar 27 11:24 root/
drwxr-xr-x  4 root root 4096 Feb 17 02:09 run/
lrwxrwxrwx  1 root root   8 Apr 22  2024 sbin -> usr/sbin/
drwxr-xr-x  2 root root 4096 Feb 17 02:02 srv/
dr-xr-xr-x 13 root root   0 Mar 27 11:20 sys/
drwxrwxrwt  1 root root 4096 Mar 26 20:02 tmp/
drwxr-xr-x  1 root root 4096 Feb 17 02:02 usr/
drwxr-xr-x  1 root root 4096 Feb 17 02:09 var/
root@d8c4820eaf12:/# cd FLM-Volume/
root@d8c4820eaf12:/FLM-Volume# ll
total 8
drwxr-xr-x 2 root root 4096 Mar 26 19:54 ./
drwxr-xr-x 1 root root 4096 Mar 27 11:39 ../
root@d8c4820eaf12:/FLM-Volume# █

```

after enter into the container3 only container files you will see not volume
when you go to volume folder it is empty



Next point is even if we stop container or remove we can still access the volume

Key Point: Images $\neq$ Volumes

Volumes are independent!

to check container which volume is attached i.e : `docker inspect cont1`

there you can get volume details and volume path local and container

to see list of volumes: `docker volume ls`

docker default path is : `/var/lib/docker`

`docker stop harish1 harish2`

`docker container prune`

`cd /var/lib/docker`

```
harish@aold2402006:~$ sudo -s
root@aold2402006:/home/harish# cd /var/lib/docker
root@aold2402006:/var/lib/docker# ll
total 56
drwx--x--- 12 root root 4096 Mar 27 08:30 ./
drwxr-xr-x 78 root root 4096 Mar 24 21:01 ../
drwx--x--x  3 root root 4096 Feb 10 11:54 buildkit/
drwx--x---  5 root root 4096 Mar 18 01:15 containers/
-rw-----  1 root root   36 Feb 10 11:54 engine-id
drwx-----  3 root root 4096 Feb 10 11:54 image/
drwxr-xr-x  3 root root 4096 Feb 10 11:54 network/
-rwxr-xr-x  1 root root 1572 Feb 10 11:59 nuke-graph-directory.sh*
drwx--x--- 25 root root 4096 Mar 27 08:30 overlay2/
drwx-----  3 root root 4096 Feb 10 11:54 plugins/
drwx-----  2 root root 4096 Mar 27 08:30 runtimes/
drwx-----  2 root root 4096 Feb 10 11:54 swarm/
drwx-----  2 root root 4096 Mar 27 08:30 tmp/
drwx-----x 2 root root 4096 Mar 27 08:30 volumes/
root@aold2402006:/var/lib/docker#
```

now you can see here containers , images , volumes , so etc

`cd volumes`

`cd goto our volume`

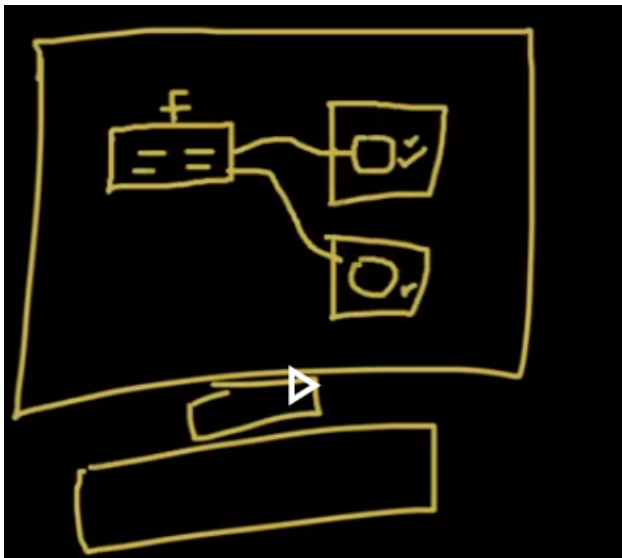
`cd _data/`

then here what we have created evrything we can see and container are removed

How to crate volumes mounting

data sharing till now happen container to container

host to container how to replicate host : means where you run data



If you update the folder attached containers will update automatically will see
 mkdir myapp
 this folder make as a host and how to give
 create a container

```
docker run -it --name cont-one -v /root/myapp:/Mustafa-volume --privileged=true ubuntu bash
```


![375cdcf4658ceb41ffad9d3a5c65a207.png]

(_resources/3d31fb34ade742a7b0c4bfc7d66d09c8.png)
 here inside container volume i have created 3 files then i exit and
 came back to myapp folder then here also i am able to see 3 files
 outside host one file i put the content

vim azure

jakdfkjkejkjrj

enter to container

docker attach cont-one

![cf6314fd55c2478419ca11d9f17f099d.png]

(_resources/a18e1e70a94744adb5032dcb151bbe66.png)

container also data is updated

cont1 lo unna same data cont2 ki vellali same command

```
docker run -it --name cont-two -v /root/myapp:/harish-volume --privileged=true ubuntu bash
```

![e7e0f21384bce655dd70a03d44df659d.png]

(_resources/f62aa607773e4f9f9f8dac915c9ed30d.png)

here container 2 also come same files

we have alternative command also

```
docker run -it --name cont-two -v \$(pwd):/harish-volume --privileged=true ubuntu bash
```



```

<br/>docker volume ls : list of volume
doekr volume rm volume_name : to delete volume
Note: if the volume is attach to container then we can not delete
docker volume prune : not using volumes will delete
docker volume create harish : to create own volume without
container
docker volume inspect volume_name
<br/>while create container i assign volume
Now separate create volume one i will keep all files and new
created container can we keep : yes
will see steps: this we can say mounting volume
Mounting
<br/>docker volume create paytm
if i want to keep data here so here i know path
/var/lib/docker/volumes/paytm inside here goto \_data
here i created 5 files i.e touch paytm-files{1..5}
<br/>![742272bdcabc112d3af377ebecblea995.png]
(_resources/90f4891016bd4364af7f7860b6d65192.png)
<br/>here as above pic docker installed by snap thats way other
default path id /var/lib/docker
<br/>so paytm ane volume ni container ki mount cheyali
docker run -itd --name cont99 --mount source=paytm,destination=/my-
volume ubuntu
![eea8cc3fbf14682f4420df6548c68050.png]
(_resources/0a8c0514b23e4c35aac9fe088dda6dfd.png)
<br/>so here my-volume is mounted
cd my-volume
we can see the files as we created
<br/>now again we will do
docker volume create zomato
if i want to keep data where to go
cd /var/lib/docker
cd volumes/zomoto/\_data
vim index.html
<h1> hey this is harish</h1>
<br/>i want to use nginx this so
docker run -it --name zomato --mount
source=zomato,destination=/usr/share/nginx/html -p 8089:80 nginx
<br/>copy ip adresspaste with url
if i enter new data just need to do refresh new data also it is not
come
we can do as repos we did
<br/><br/><br/>So this is one process

```

[illegible]